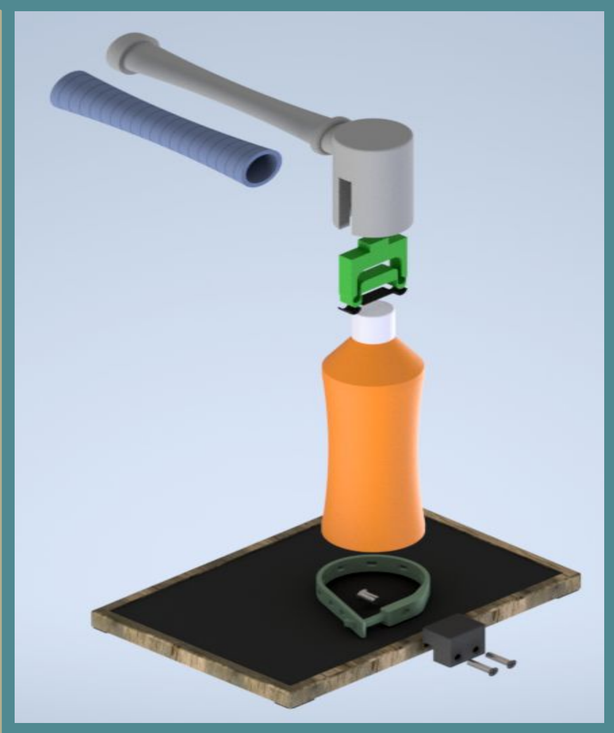
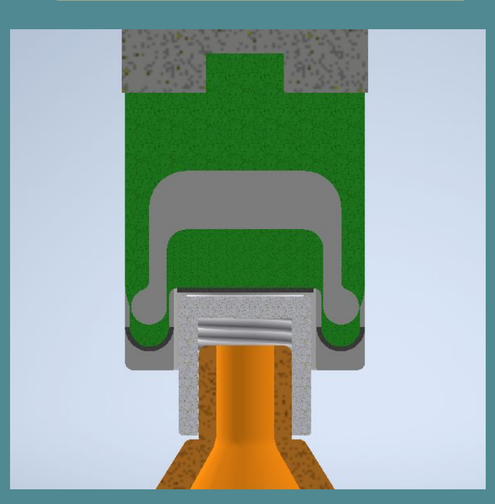
Compliant Mechanism Child Lock Opener: Production and Environmental Impact

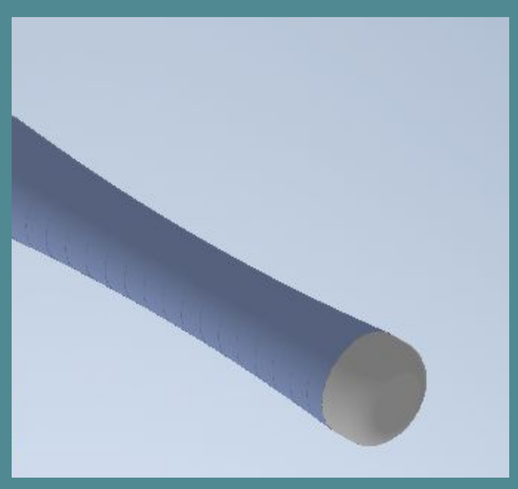
Component	Material Requirements	Material Choice	Manufacturing Process
Compliant Mechanism	Flexible, Easy to deform but does not fracture, not very stiff. Must also be food safe.	Polyethylene (PE) for the compliant mechanism. Rubber mechanism for the surrounding.	Injection Moulding
Handle	Soft, flexible, and ergonomic material for the handle sleeve. The main handle should be strong and lightweight.	Ethylene vinyl acetate (EVA) for the ergonomic sleeve. Aluminium for the handle.	Injection moulding for the sleeve. Sand Casting for the handle.
Base	Anti-slip material to ensure the bottle does not slip. Lightweight to not cause injuries if dropped.	The case is made of lightweight bamboo wood. The belt and base are both made of rubber.	Injection Moulding for the rubber Resin impregnation and curing for the wooden case



Compliant Mechanism



Sleeve for Handle



Why Polyethylene (PE)?

PE was chosen to make the compliant mechanism due to its characteristics, which followed the design requirements. The object is aimed to be produced in large batches; this thermoplastic is cheap, widely recyclable, and easy to alter its molecular structure; therefore, it is ideal for producing the complex shape the compliant mechanism requires. It is also highly flexible, allowing it to absorb the energy and reduce the overall stiffness of the system, the characteristics that a complaint mechanism requires. Furthermore, PE has a lower yield strength of 18 MPa as compared to its plastic countermart, Polypropylene (PP) having a yield strength of 21 MPa, thus making it ideal for our target audience. PE is also corrosion-resistance, durable and 100% recyclable, as they withstand up to 10 heating cycles due to their simple structure, which makes it ideal for the application. Moreover, the main reason PE was chosen over PP was due to its higher fatigue strength; with PP having a fatigue strength of 11-17 MPa, and PE having a fatigue strength of 21-23 MPa.

Why injection Moulding?

PE resists to high shear forces, which occurs when the plastic is melted and when the mould is being filled with the melt. PE has also Pseudoplastic behavior, which is more suitable for injection moulding, PE has also a low softening temperature (around 80º) and flow well in soft state,which makes it ideal for injection moulding.

Why Ethylene vinyl acetate (EVA)?

EVA is used for the grip on the handle, therefore a comfortable, skin- harmless material is required. EVA is hard lightweight copolymer resin with good foamability performance, which is ideal for the handle. EVA is also easy to recycle and cheaper than most rubber materials. It also has smooth surfaces which makes it more pleasant for the user.

Why injection Moulding?

Injection moulding with EVA produces high precise products. EVA (250º) has a low melting point, therefore costs can be significantly reduced as low energy requirements and low maintenance operations are required. It also allows batch production, which is essential for the product.

Redesign?

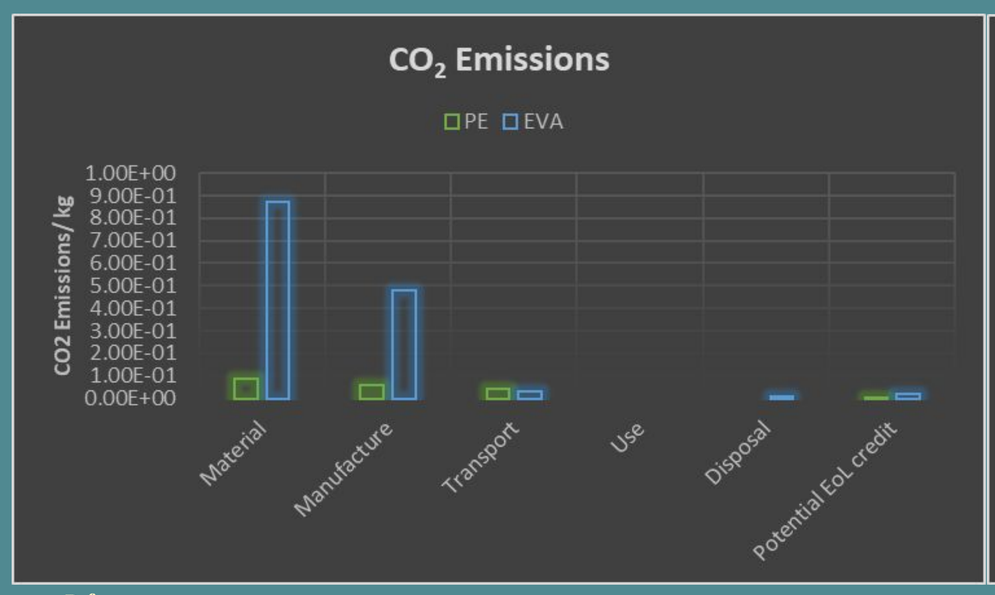
From the eco audit it can be seen that EVA has a higher carbon footprint and energy consumption. Furthermore, EVA is harder to recycle, thus a bio-based EVA can be used that has the same properties but is made of renewable raw materials. This would also make the disposal process easier as the entire component could be recycled. The manufacturing process for the EVA could be changed to extrusion, as it requires less energy, and has a lower carbon footprint, as it emits 5.7 MJ/kg of energy wheres, moulding emits 17.2 MJ/kg.

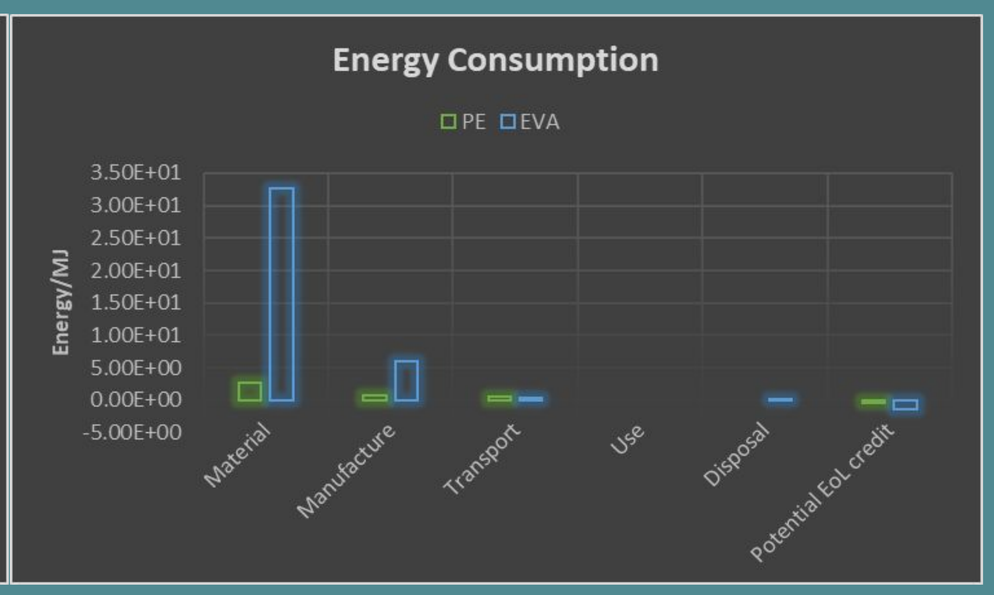
Both PE and EVA have an end of life debit for the CO_2 emissions, and only has a credit for the EVA energy consumption, thus,, the components can be designed to be re-used for other purposes.

Assumptions for Eco-Audit

1. The components are sourced from the nearest manufacturing sites to Bath. LDPE is sourced from “Fern Plastics” and is 187 km away. The nearest EVA plant is “Cinek Chemicals” and is 109.5 km away. They are both assumed to be transported by 32 metric ton trucks.
2. 100% of the LDPE is assumed to be recycled, whereas, only 10% of the EVA is assumed to be recycled due to its limited recycling scope.
3. It is assumed that the mechanism does not consume any energy during its usage and storage, thus has a zero carbon footprint in this regard.

Eco-Audit for both materials





Conclusion

1. The material phase contributes to the most carbon emissions and energy consumption for both PE and EVA.
2. Only EVA has a potential end of life debit or the energy consumption, whereas, everything else is in credit.
3. EVA is mostly dumped into landfills as it is difficult to recycle, thus showing a rise in the disposal phase.
4. The EVA manufacturing phase also contributes highly to the carbon footprint.

References

1. Ashby, M. (2013). Materials and the environment : Eco-informed material choice (2nd ed.). Amsterdam ; London: Butterworth-Heinemann.
2. Bio-based Ethylene vinyl acetate (EVA) [Internet]. Available from: https://fkur.com/en/bioplastics/im-green-eva/
3. Injection Moulding of Polyethelyne [Online]https://www.cavitymold.com/injection-molding-of-polyethylene/
4. RUBBER VS. EVA STABLE MATHS , 28 Feb 2017,[online]https://www.mats4u.co.uk/blog/rubber-vs-eva-stable-mats.html#:~:text=Cost%20%26%20Value,percentage%20of%20natural%2C%20costly%20materials
5. Ethylene Vinyl Acetate (EVA) Recycling [online].https://www.wastetrade.com/resources/introduction-to-plastics/types-of-plastics/ethylene-vinyl-acetate-eva-recycling/#:~:text=EVA%20Recyclability,used%20to%20make%20new%20products.
6. Low Density Polyethylene (LDPE) 29th April 2022, [online] https://www.xometry.com/resources/materials/low-density-polyethylene-ldpe/
7. Bamboo Board - an overview | ScienceDirect Topics [Internet]. www.sciencedirect.com. Available from: https://www.sciencedirect.com/topics/engineering/bamboo-board#:~:text=Bamboo%20board%20is%20made%20from